

MERN Stack Integration Assignment

1. React Frontend Sends Playlist Data to Node Backend and MongoDB

Backend (server.js)

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();

app.use(cors());
app.use(express.json());

mongoose.connect("mongodb://127.0.0.1:27017/musicdb");

const PlaylistSchema = new mongoose.Schema({
    name: String
});

const Playlist = mongoose.model("Playlist", PlaylistSchema);

app.post("/playlist", async (req, res) => {
    try {
        const playlist = new Playlist({
            name: req.body.name
        });

        await playlist.save();
    }
});
```

```
    res.status(200).json({
      message: "Playlist saved successfully"
    });

  } catch (error) {
    res.status(500).json({
      error: error.message
    });
  }
});
```

```
app.listen(5000);
```

React Frontend

```
import { useState } from "react";
```

```
function App() {
```

```
  const [name, setName] = useState("");
```

```
  const submitPlaylist = async (e) => {
    e.preventDefault();
```

```
    const response = await fetch("http://localhost:5000/playlist", {
      method: "POST",
      headers: {
        "Content-Type": "application/json"
```

```
    },  
    body: JSON.stringify({ name })  
  });  
  
  const data = await response.json();  
  
  alert(data.message);  
};  
  
return (  
  <form onSubmit={submitPlaylist}>  
    <input  
      type="text"  
      placeholder="Playlist Name"  
      value={name}  
      onChange={(e) => setName(e.target.value)}  
    />  
    <button type="submit">Save Playlist</button>  
  </form>  
);  
}  
  
export default App;
```

2. Add Restaurant Form with MongoDB Storage

Backend

```
const RestaurantSchema = new mongoose.Schema({  
  name: String,  
  cuisine: String,  
  rating: Number  
});
```

```
const Restaurant = mongoose.model(  
  "Restaurant",  
  RestaurantSchema  
);
```

```
app.post("/restaurants", async (req, res) => {  
  
  try {  
  
    const restaurant = new Restaurant(req.body);  
  
    const savedRestaurant =  
      await restaurant.save();  
  
    res.status(201).json(savedRestaurant);  
  
  } catch (error) {  
  
    res.status(500).json({
```

```
        error: error.message
    });

    }
});

React Frontend

import { useState } from "react";

function RestaurantForm() {

    const [restaurant, setRestaurant] =
        useState(null);

    const submitRestaurant = async (e) => {

        e.preventDefault();

        const response = await fetch(
            "http://localhost:5000/restaurants",
            {
                method: "POST",
                headers: {
                    "Content-Type":
                        "application/json"
                },
                body: JSON.stringify({
                    name: "Pizza Hub",
```

```

        cuisine: "Italian",
        rating: 4.5
    })
}
);

const data = await response.json();

setRestaurant(data);
};

return (
    <div>
        <button onClick={submitRestaurant}>
            Add Restaurant
        </button>

        {restaurant && (
            <div>
                <h3>{restaurant.name}</h3>
                <p>{restaurant.cuisine}</p>
                <p>{restaurant.rating}</p>
            </div>
        )}
    </div>
);
}

```

```
export default RestaurantForm;
```

3. GET /movies Endpoint and React Display

Backend

```
const MovieSchema = new mongoose.Schema({  
  title: String,  
  year: Number  
});
```

```
const Movie = mongoose.model(  
  "Movie",  
  MovieSchema  
);
```

```
app.get("/movies", async (req, res) => {
```

```
  try {
```

```
    const movies =  
      await Movie.find();
```

```
    res.json(movies);
```

```
  } catch (error) {
```

```
    res.status(500).json({
```

```
        error: error.message
    });

    }
});

React Frontend

import { useEffect, useState } from "react";

function Movies() {

    const [movies, setMovies] =
        useState([]);

    useEffect(() => {

        fetch("http://localhost:5000/movies")
            .then(res => res.json())
            .then(data => setMovies(data));

    }, []);

    return (
        <div>
            <h2>Movie List</h2>

            <ul>
                {movies.map(movie => (
```



```
        <li key={movie._id}>
          {movie.title}
        </li>
      )}
    </ul>
  </div>
);
}
```

```
export default Movies;
```

4. Error Handling in MERN Flow

Backend

```
app.post("/restaurants", async (req, res) => {

  try {

    const restaurant =
      new Restaurant(req.body);

    const savedRestaurant =
      await restaurant.save();

    res.status(201).json(savedRestaurant);

  } catch (error) {
```

```
    res.status(500).json({
      error:
        "Database operation failed"
    });

  }
});
```

Frontend

```
const submitRestaurant = async () => {

  try {

    const response = await fetch(
      "http://localhost:5000/restaurants",
      {
        method: "POST",
        headers: {
          "Content-Type":
            "application/json"
        },
        body: JSON.stringify({
          name: "Pizza Hub",
          cuisine: "Italian",
          rating: 4.5
        })
      }
    );
```

```
const data =  
  await response.json();  
  
  if (!response.ok) {  
    throw new Error(data.error);  
  }  
  
  alert("Restaurant Saved");  
  
} catch (error) {  
  
  alert(error.message);  
  
}  
};
```

5. Async/Await Refactoring

Before

Movie.find()

```
.then(data => {  
  res.json(data);  
})  
.catch(error => {  
  res.status(500).json({  
    error: error.message  
  });  
});
```

```
});
```

After

```
app.get("/movies", async (req, res) => {
```

```
  try {
```

```
    const movies =
```

```
      await Movie.find();
```

```
    res.json(movies);
```

```
  } catch (error) {
```

```
    res.status(500).json({
```

```
      error: error.message
```

```
    });
```

```
  }
```

```
});
```

MongoDB Connection Using Async/Await

```
async function connectDB() {
```

```
  try {
```

```
    await mongoose.connect(
```

```
      "mongodb://127.0.0.1:27017/moviedb"
```

```
    );
```

```
    console.log(  
        "MongoDB Connected"  
    );  
  
} catch (error) {  
  
    console.log(  
        "Database Connection Failed"  
    );  
  
    console.log(error.message);  
  
}  
}  
  
connectDB();
```